

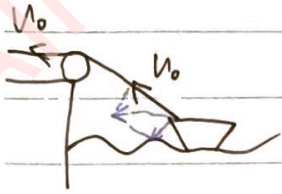
§1. 质点运动学.

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1. 运动方程: $\vec{r} = \vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$.
(位置矢量表达式)

2. 轨道方程: (消去 t). 需 x, y, z 的关系式.



运动方程: 从实际位置出发
方向方程: 从效果出发.

eg. 质点作一维运动. $a = -kx$. k 为正常数. $t=0$ 时. $x=x_0$. 求运动规律.

$$a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx} = -kx$$

$$\Rightarrow \int_0^v v dv = \int_{x_0}^x (-kx) dx$$

$$v = \frac{dx}{dt} = \sqrt{k(x_0^2 - x^2)}$$

$$\Rightarrow \int_{x_0}^x \frac{dx}{\sqrt{x_0^2 - x^2}} = \int_0^t \sqrt{k} dt$$

$$\arcsin \frac{x}{x_0} \Big|_{x_0}^x = \sqrt{k} t$$

$$\Rightarrow x = x_0 \cdot \cos(\sqrt{k} t).$$

$\therefore$ 质点沿 x 轴做简谐振动.

e.g. 气球. $v_y = 1 \text{ m/s}$. $v_x = 2y \text{ m/s}$. 求 (1) 运动方程 (2) 加速度. (3) a_n, a_t .
 $y = t^2 \text{ m}$ $v_x = 2t \text{ m/s}$ $x = t^2 \text{ m}$

$\vec{r} = t^2 \vec{i} + t \vec{j}$ ($\vec{r} = x(t)\vec{i} + y(t)\vec{j}$)

$$(2) \vec{v} = \frac{d\vec{r}}{dt} = 2t\vec{i} + \vec{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = 2\vec{i}$$

$$(3) a_t = \frac{dv}{dt} = \frac{d\sqrt{4y^2+1}}{dt} = \frac{4y}{\sqrt{4y^2+1}}$$

$$a_n = \sqrt{a^2 - a_t^2} = \frac{2}{\sqrt{4y^2+1}}$$

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§2. 運動と力.

$$1. \text{ 4} =: \vec{F} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = \frac{dm}{dt} \vec{v} + \frac{d\vec{v}}{dt} m = m \cdot \frac{d\vec{v}}{dt} = m\vec{a}.$$

第二章. 运动与力.

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一. 牛顿三大定律

1. 牛一: 小惯性定律.

成立: 小惯性系 (地面 S_A 参考系)

不成立: 非... (如 S_B ...)

小惯性力 (假想力)



$$\begin{aligned} 2. \text{牛二: } \vec{F} &= \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} \\ &= \frac{dm}{dt} \cdot \vec{v} + \frac{d\vec{v}}{dt} \cdot m \\ &= m \cdot \frac{d\vec{v}}{dt} \\ &= m\vec{a} \end{aligned}$$

$$\vec{F} + \vec{F}_i = m\vec{a}$$

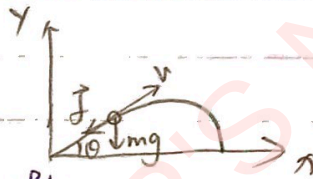


3. 牛三: 力与反作用力.

二. 例题.

eg. 1. 斜抛. 空气阻力 $\vec{f} = -k m \vec{v}$. 求 t 时. 速度. 位置. 轨迹.

$$\begin{aligned} \text{沿 } x \text{ 轴: } -k m v_x &= m \frac{dv_x}{dt} \\ \text{沿 } y \text{ 轴: } -k m v_y - mg &= m \frac{dv_y}{dt} \end{aligned}$$



$$\int_0^t -k dt = \int_{v_{0x}}^{v_x} \frac{dv_x}{v_x} \Rightarrow v_x = v_0 e^{-kt} \cos \theta$$

$$\int_0^t -k dt = \int_{v_{0y}}^{v_y} \frac{dv_y}{v_y + \frac{g}{k}} \Rightarrow v_y = (v_0 \sin \theta + \frac{g}{k}) e^{-kt} - \frac{g}{k}$$

$$v = \sqrt{v_x^2 + v_y^2}$$

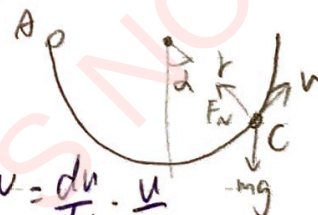
$$x = \int_0^t v_x dt = \frac{v_0 \cos \theta}{k} (1 - e^{-kt})$$

$$y = \int_0^t v_y dt = \frac{v_0 \sin \theta + \frac{g}{k}}{k} (1 - e^{-kt}) - \frac{gt}{k}$$

$$\text{轨迹: } y = \frac{v_0 \sin \theta + \frac{g}{k}}{v_0 \cos \theta} x + \frac{g}{k^2} \ln \left[1 - \frac{kx}{v_0 \cos \theta} \right]$$

eg. 2. 质量为 m 的球. 从 A 沿光滑圆轨滑下. 求 C 点 ω . F_N .

$$\begin{aligned} \text{法向: } F_N - mg \cos \alpha &= m a_n = m \frac{v^2}{r} \\ \text{切向: } mg \sin \alpha &= m a_t = m \frac{dv}{dt} \end{aligned}$$



$$\int_0^{\frac{\pi}{2}} r g \sin \alpha d\alpha = \int_0^v v dv = \frac{v^2}{2}$$

$$r g \cos \alpha = \frac{v^2}{r}$$

$$\omega = \frac{v}{r} = \pm \sqrt{2 g \cos \alpha}$$

$$\rightarrow F_N = -F_N = -3 m g \cos \alpha$$

eg. 3. 飞机制动. $m = 1.0 \times 10^3 \text{ kg}$. $v_0 = 55 \text{ m/s}$. $f = \alpha t$. $\alpha = 5 \times 10^2 \text{ N/s}$.

求: (1) 10s 后 v . (2) 10s 内滑行距离 s .

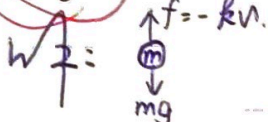
解: (1) $a = \frac{f}{m} = \frac{\alpha t}{m}$

$$\int_{v_0}^v v dv = \int_0^t \frac{\alpha t}{m} dt \Rightarrow v = v_0 - \frac{\alpha}{2m} t^2 \Rightarrow v = 30 \text{ m/s}$$

(2) $\int_{x_0}^x dx = \int_0^t v dt$

$$s = x - x_0 = v_0 t - \frac{\alpha}{6m} t^3 = 467 \text{ m}$$

eg. 4. m 下落, $f = -kv$. 求运动方程. $x(t)$.



$$mg - kv = ma = m \frac{dv}{dt}$$

$$\int_0^t dt = \int_0^v \frac{m}{mg - kv} dv$$

$$\Rightarrow t = \frac{m}{-k} \ln(mg - kv)$$

$$\Rightarrow v = \frac{mg}{k} (1 - e^{-\frac{kt}{m}})$$

$$x = \int_0^t v dt = \frac{mg}{k} \left(\frac{m}{k} e^{-\frac{k}{m}t} + t - \frac{m}{k} \right)$$

eg. 5. 求滑块 a_0 与物体相对斜面 v .

解: $F_{N2} \sin \theta = m_1 a_0$

$$m_2 a_0 + F_{N2} \sin \theta = m_2 a'_x$$

$$F_{N2} \cos \theta - m_2 g = -m_2 a'_y$$

$$\Rightarrow a_0 = \frac{m_2 \cos \theta \sin \theta}{m_1 + m_2 \sin^2 \theta} g$$

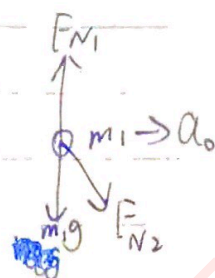
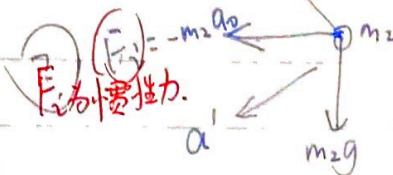
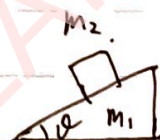
木板对地面:

$$a' = \frac{(m_1 + m_2) \sin \theta}{m_1 + m_2 \sin^2 \theta} g$$

物体对木板:

物体对地面. $a_x = a' \cos \theta - a_0$

$$a_y = a' \sin \theta$$



$$\begin{cases} F_{N2} \sin \theta = m_2 (a'_x - a_0) \\ F_{N2} \cos \theta - m_2 g = -m_2 a'_y \end{cases}$$

第三章. 动量与角动量.

一. 动量.

1. 动量: $\vec{p} = m\vec{v}$

冲量: $\vec{I} = \int_{t_0}^t \vec{F} dt$

2. 动量定理: $\vec{I} = \vec{p} - \vec{p}_0 = \vec{F}_{\text{合}} t$

质点系中, $\int_{t_0}^t \sum_{i=1}^n \vec{F}_i dt = \sum_{i=1}^n \vec{p}_i - \sum_{i=1}^n \vec{p}_{i0}$

3. 动量守恒定律: 当 $\vec{F} = \sum_{i=1}^n \vec{F}_i = 0$ 时, $\sum_{i=1}^n \vec{p}_i = \sum_{i=1}^n m_i \vec{v}_i$

if $\vec{F}_{\text{合}}$ 沿某全特方向为0, 则系统的 $\vec{p}_{\text{合}}$ 沿该方向的另量守恒.

eg.

(1) 受外力矩: 力矩和为0
 $\Rightarrow$ 角动量守恒.

静止始, $L = L_0 = 0$

$m_1 v_1 R - m_2 v_2 R = 0$

if $m_1 = m_2 \Rightarrow v_1 = v_2$

(2) $m_1 \neq m_2$, 角动量(不)守恒

$L = m_1 v_1 R - m_2 v_2 R$

$M_{\text{合}} = \frac{dL}{dt} = (m_2 - m_1)gR$

if $m_2 > m_1 \Rightarrow > 0$

$\therefore L_0 = 0$

$\therefore L > 0$

$\Rightarrow v_1 > v_2$

同理, if $m_2 < m_1, v_2 > v_1$

二. 角动量.

1. 力是改变质点运动状态的原因.

力矩是改变刚体转动状态的原因.

力臂: d

力矩: $\vec{M} = \vec{r} \times \vec{F}$

大小: $M = F r \sin \theta$

方向: 右手定则.

$\frac{d\vec{L}}{dt} = \frac{d(\vec{r} \times \vec{p})}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \frac{d\vec{p}}{dt} \times \vec{r}$

$= \vec{v} \times \vec{p} + \vec{F} \times \vec{r}$

$= \vec{F} \times \vec{r}$

$= \vec{M}$

2. 角动量: $\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$ 大小: $L = r m v \sin \theta$

角动量定理: $\int_{t_1}^{t_2} \vec{M} dt = \vec{L}_2 - \vec{L}_1$

冲量矩.

质点系中, $\sum \vec{M}_i = \sum \frac{d\vec{L}_i}{dt} = \frac{d\vec{L}}{dt}$

3. 角动量守恒定律:

由于合外力矩为0, 物体角动量守恒: $\vec{L} = \sum m_i \vec{r}_i \vec{\omega}$

eg. 长 l , 线密度 λ , 绳从口落. 求 $v(x)$.

动量定理:

$F = m \cdot g = \lambda y g$

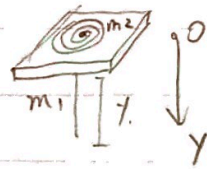
$\vec{F} dt = \lambda y g dt = d\vec{p} = d\lambda y v = \lambda dy v$

$y g = \frac{dy v}{dt} \frac{dy}{dy}$

$y g dy = d(y v) \cdot v$

$\int_0^y y^2 g dy = \int_0^y y v dy$

$\Rightarrow \frac{y^3}{3} g = \frac{v^2}{2} \Rightarrow v = \sqrt{\frac{2 y g}{3}}$



第四章. 功和能.

一. 功.

$$1. W = \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r} = \int F_x dx + \int F_y dy + \int F_z dz.$$

合力功 == 各分力功之和

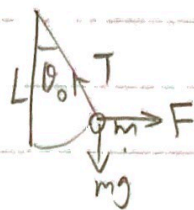
2. 功率 < 平均: $\bar{P} = \frac{\Delta W}{\Delta t}$

瞬时: $P = \frac{dW}{dt} = \frac{d(\vec{F} \cdot \vec{r})}{dt} = \frac{d\vec{F}}{dt} \cdot \vec{r} + \frac{d\vec{r}}{dt} \cdot \vec{F}$

$$= \vec{F} \cdot \vec{v}$$

eg. 水平力 F 作用于 m . 缓慢上升. $0 \rightarrow \theta_0$.

$$\begin{cases} F - T \sin \theta = 0 \\ T \cos \theta - mg = 0 \end{cases} \Rightarrow F = mg \tan \theta$$



$$dW = \vec{F} \cdot d\vec{s} = F \cdot ds \cdot \cos \theta$$

$$= F \cdot L \cos \theta \cdot d\theta$$

$$= mg L \sin \theta \cdot d\theta$$

$$W = mgL (1 - \cos \theta_0)$$

3. 动能定理

$$dW = \vec{F}_{\text{合}} \cdot d\vec{r} = m \frac{d\vec{v}}{dt} \cdot d\vec{r} = m \vec{v} \cdot d\vec{v}$$

$$= m v_x dv_x + m v_y dv_y + m v_z dv_z = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2)$$

$$= \frac{1}{2} m dv^2 = m v dv$$

$$W = \int dW = \int_{v_0}^v m v dv = \frac{1}{2} m v^2 - \frac{1}{2} m v_0^2$$

$$\int \vec{F} \cdot d\vec{s} = \frac{1}{2} m v^2 - \frac{1}{2} m v_0^2 \quad \text{力的空间累积效应}$$

$$\int \vec{F} \cdot d\vec{t} = m \vec{v} - m \vec{v}_0 \quad \text{力的时间累积效应}$$

eg. $m = 4 \text{ kg}$. $\vec{F} = 2xy\vec{i} + 3x^2\vec{j}$ (SI). 沿 $x^2 = 9y$. 从 $(0,0) \rightarrow (3,1)$. 求 W .

$$W = \int_0^Q \vec{F} \cdot d\vec{r}$$

$$= \int_0^Q (F_x dx + F_y dy)$$

$$= \int_0^Q (2xy dx + 3x^2 dy)$$

将 $x^2 = 9y$ 代入.

$$= \int_0^3 \frac{2}{9} x^3 dx + \int_0^1 27y dy = 18 \text{ J}$$

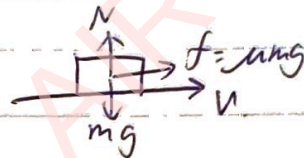
$$W = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2$$

$$\rightarrow v_2 = 3 \text{ m/s}$$



eg. m 放在传送带上, μ, v 求: (1) 何时变速; (2) 相对地面 m 位移.

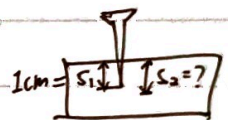
分析: (1) 动量定理: $F_f t = mv - 0$
 $\rightarrow t = \frac{v}{\mu g}$



(2) 动能定理: $W = F_f s = \frac{1}{2}mv^2 - 0$
 $\rightarrow s = \frac{v^2}{2\mu g}$

eg. m_0 铁锤, m 钉, 木板阻力 $-kx$, 两次锤子均 v_0 , 求 s_2 .

分析: $m_0 v_0 = (m + m_0) v_1$ (共速)
 $\rightarrow v_1 = \frac{m_0 v_0}{m + m_0} (\approx v_0)$ 不够也能算. $\rightarrow$ 克服阻力做功.



$0 - \frac{1}{2}mv_0^2 = \int_0^{s_1} -kx dx$
 $0 - \frac{1}{2}mv_0^2 = \int_{s_1}^{s_1+s_2} -kx dx \Rightarrow s_2 = (\sqrt{2} - 1) \text{ cm}$

e.g. 水平方向, 动量守恒:

$m v_0 = (m + m_0) v_B$ ①

系统, 机械能守恒:

$\frac{1}{2}(m + m_0)v_A^2 = \frac{1}{2}(m + m_0)v_B^2 + \frac{1}{2}k(L - L_0)^2$ ②

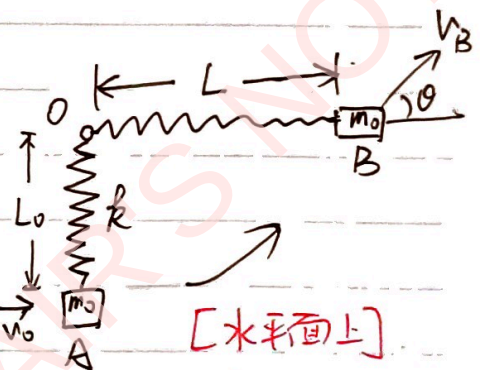
由 ①, ②, 得:

$v_B = \left[\frac{m^2 v_0^2}{(m + m_0)^2} - \frac{k(L - L_0)^2}{m + m_0} \right]^{\frac{1}{2}}$

角动量守恒:

$(m + m_0)v_A L_0 = (m + m_0)v_B L \sin \theta$

$\Rightarrow \theta = \arcsin \frac{m L_0 v_0}{L \left[\frac{m^2 v_0^2}{m + m_0} - k(L - L_0)^2 \right]^{\frac{1}{2}}}$



第五章 刚体.

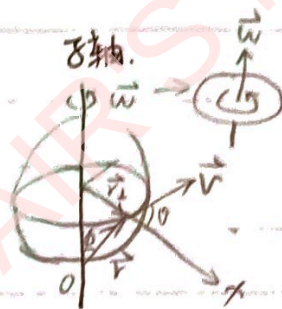
1. $\vec{v} = \vec{\omega} \times \vec{r}$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v}$$

切向加速度 向轴加速度

$v = r_{\perp} \omega$

$a_n = r_{\perp} \omega^2$ $a_t = \frac{dv}{dt} = r_{\perp} \alpha$

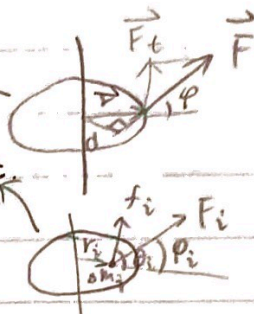


$$\begin{cases} \omega = \omega_0 + \alpha t \\ \theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2 \\ \omega^2 - \omega_0^2 = 2\alpha(\theta - \theta_0) \end{cases}$$

2. $\vec{M} = \vec{r} \times \vec{F} = \vec{r} \cdot d\vec{F} = \vec{r} \cdot \vec{F}_t$

对任意 Δm_i : $f_i \sin \theta_i + F_i \sin \phi_i = \Delta m_i a_{it}$

~~$f_i \cos \theta_i - F_i \cos \phi_i = \Delta m_i \omega^2 r_i$~~



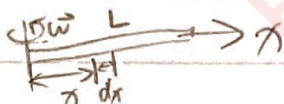
$$\begin{aligned} \sum r_i f_i \sin \theta_i + \sum r_i F_i \sin \phi_i &= \sum r_i \Delta m_i a_{it} \\ \text{内力矩为0} \quad \sum F_i \times r_i &= \sum \Delta m_i r_i^2 \alpha \\ &= \vec{M} = J \alpha \end{aligned}$$

$\Rightarrow$ 转动定律: $\vec{M} = J \cdot \vec{\alpha}$

$J = \sum \Delta m_i \cdot r_i^2$ (分点)
 $= \int r_i^2 \cdot dm$ (连续)

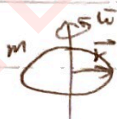
J 转动惯量 (书 P81, 82)

eg. 求 m, L 杆转动惯量



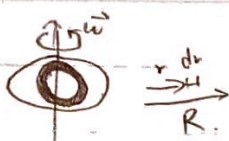
解: $J = \int_0^L x^2 \frac{dm}{dx} = \frac{m}{L} \int_0^L x^2 dx = \frac{mL^2}{3}$

eg. 求圆环的 J .



解: $J = \int r^2 dm = r^2 \cdot m$

eg. 求 m, R 圆盘的 J .

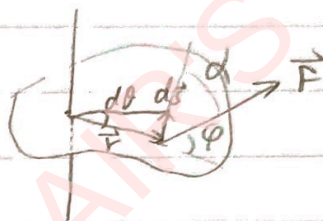


解: $J = \int_0^R r^2 \frac{dm}{dr} = \frac{mR^2}{2}$

$m \cdot \frac{2\pi r \cdot dr}{\pi R^2}$

二. 功和能关系

$$\begin{aligned} 1. \text{ 功: } dW &= \vec{F} \cdot d\vec{s} = F ds \cos\alpha \\ &= F \cdot r d\theta \sin\varphi \\ &= (F \cdot r \sin\varphi) \cdot d\theta \\ &= M \cdot d\theta \end{aligned}$$



$$W = \int M d\theta$$

$$P = \frac{dW}{dt} = \frac{M d\theta}{dt} = M\omega, \quad (\text{恒 } M, W = M\theta)$$

2. 动能定理

$$M = J\alpha = J \frac{d\omega}{dt} = J \frac{d\omega}{d\theta} \frac{d\theta}{dt} = J \frac{d\omega}{d\theta} \cdot \omega$$

$$\Rightarrow M d\theta = J \omega d\omega$$

$$\Rightarrow \int_{\theta_1}^{\theta_2} M d\theta = J \int_{\omega_1}^{\omega_2} \omega d\omega$$

$$\Rightarrow \int_{\theta_1}^{\theta_2} M d\theta = \frac{1}{2} J \omega_2^2 - \frac{1}{2} J \omega_1^2$$

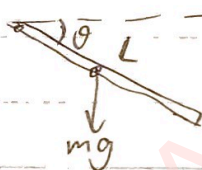
eg. 求 $\omega = \omega(\theta)$.

$$\text{例: } M = \vec{F} \times \vec{r} = mg \cdot \left(\frac{L}{2}\right) \cdot \cos\theta$$

$$\int W = \frac{1}{2} J \omega^2 - 0$$

$$W = \int M d\theta = \int_0^\theta \frac{1}{2} mg \frac{L}{2} \cos\theta d\theta = \frac{1}{2} mg L \sin\theta$$

$$\Rightarrow \omega = \sqrt{\frac{3g \sin\theta}{L}}$$



$$\text{例2: } \alpha = \frac{M}{J} = \frac{3g \cos\theta}{2L}$$

$$\alpha = \frac{d\omega}{dt} = \frac{d\omega}{d\theta} \cdot \frac{d\theta}{dt} = \omega \frac{d\omega}{d\theta}$$

$$\Rightarrow \int_0^\omega \omega d\omega = \int_0^\theta \alpha d\theta \Rightarrow \omega = \dots$$

3. 角动量定理: $\int_{t_1}^{t_2} \vec{M} dt = \vec{L}_2 - \vec{L}_1$

4. 角动量守恒定理: 当 $\sum M_i = 0$ 时, $\sum J_i \omega_i = \text{const}$

eg. 人坐在转盘上, 人+盘 J_0 , 哑铃 m . 初始 ω , 求双臂收至 $r_1 \rightarrow r_2$ 时 ω 及 ΔE .

由角动量守恒:

$$(J_0 + 2mr_1^2) \omega_1 = (J_0 + 2mr_2^2) \omega_2$$

$$\Rightarrow \omega_2 = \dots$$

$$\Delta E_k = \frac{1}{2} (J_0 + 2mr_2^2) \omega_2^2 - \frac{1}{2} (J_0 + 2mr_1^2) \omega_1^2$$

eg. 求碰撞后 ω , 最大上摆角度 θ .

碰撞过程, 角动量守恒:

$$m \frac{3}{4} L v = (J_M + J_m) \omega$$

$$J_M = \frac{1}{3} M L^2, J_m = m \left(\frac{3}{4} L\right)^2$$

$$\Rightarrow \omega = \dots$$

上摆过程, 机械能守恒:

$$\frac{1}{2} (J_M + J_m) \omega^2 = mg \frac{3}{4} L (1 - \cos \theta) + Mg \frac{L}{2} (1 - \cos \theta)$$

$$\Rightarrow \theta = \dots$$

eg. m_A, m_B 两圆盘共圆心, r_A, r_B .

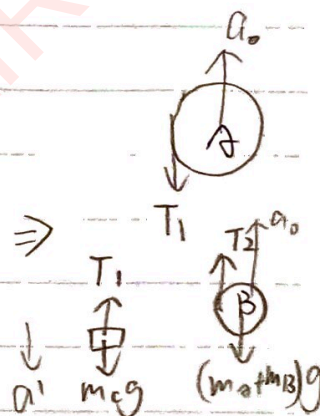
$$\begin{cases} T_2 - T_1 - (m_A + m_B)g = (m_A + m_B)a_0 \\ T_1 \cdot r_A - T_2 \cdot r_B = (J_A + J_B)\alpha = J\alpha \end{cases}$$

$$m_C g - T_1 = m_C a'$$

$$a_0 = r_B \alpha$$

$$a' = r_A \alpha - a_0 = r_A \alpha - r_B \alpha$$

$$\Rightarrow a_0 = \dots$$



第四章. 振动.

一. 简谐振动. (运动学)

1. $x(t) = A \cos(\omega t + \varphi)$

2. 特征量. 1). 振幅 A .

2). 角频率 (圆频率): $\omega = \frac{2\pi}{T}$

周期 T . 频率 $\nu = \frac{1}{T}$.

3). 初相位 $\Phi = \omega t + \varphi$

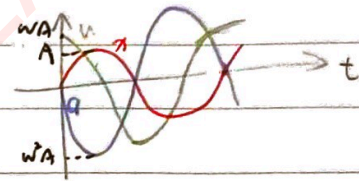
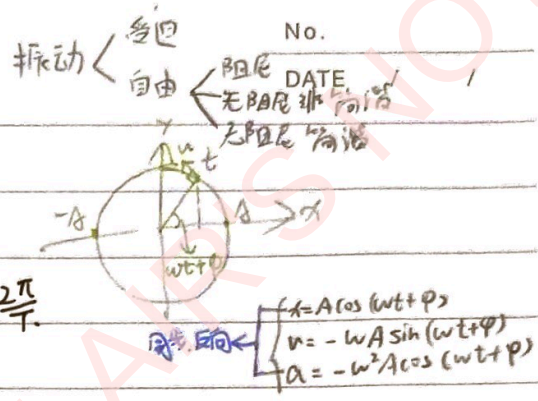
3. $v = \frac{dx}{dt}$. $a = \frac{dv}{dt}$

4. 相位差. $\Delta\Phi = (\omega_2 t + \varphi_2) - (\omega_1 t + \varphi_1)$

if $\Delta\Phi = \varphi_2 - \varphi_1 > 0 \Rightarrow$ 振动 2 超前 1; 注: $\Delta\Phi < \pi$ 作判断依据.
or ... 1 滞后 2.

if $\Delta\Phi = \pm 2k\pi \Rightarrow$ 同步

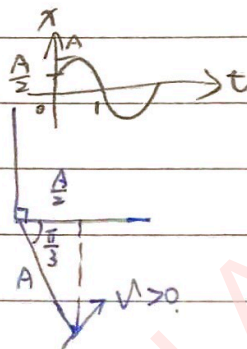
if $\Delta\Phi = \pm(2k+1)\pi \Rightarrow$ 反相



eg. - 谐振子如图. 求振动方程.

$t=0$ 时, $\begin{cases} x = \frac{A}{2} \\ v > 0 \end{cases} \Rightarrow \varphi = -\frac{\pi}{3}$

$\frac{\frac{\pi}{3} + \frac{A}{2}}{1s} = \frac{2\pi}{T} = \omega$



$\Rightarrow x = A \cos(\frac{5}{6}\pi t - \frac{\pi}{3})$

二. (动力学)

1. $\omega = \sqrt{\frac{k}{m}}$

初始: $(x_0, \frac{v_0}{\omega})$

$A = \sqrt{x_0^2 + (\frac{v_0}{\omega})^2}$

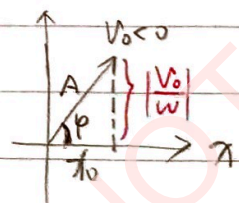
$\tan \varphi = -\frac{v_0}{\omega x_0}$

$\ddot{x} = -kx = m \frac{d^2 x}{dt^2} \Rightarrow \frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$
 $\omega = \sqrt{\frac{k}{m}}$
 $T = 2\pi \sqrt{\frac{m}{k}}$

通解: $x(t) = A e^{i(\omega t + \varphi)}$

取实部: $x(t) = A \cos(\omega t + \varphi)$

$t=0$ 时, $\begin{cases} x_0 = A \cos \varphi \\ v_0 = -\omega A \sin \varphi \end{cases}$



2. 简谐振动. 判定条件: 1. $x(t) = A \cos(\omega t + \varphi)$

2. $\frac{d^2 x}{dt^2} + \omega^2 x = 0$

3. $\vec{F} = -k\vec{x}$

No.

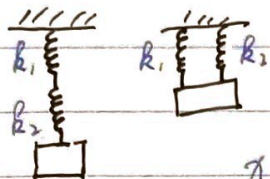
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3. 能量

1. 动能: $E_k = \frac{1}{2}mv^2 = \frac{1}{2}BA^2\sin^2(\omega t + \varphi)$

2. 势能: $E_p = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\cos^2(\omega t + \varphi)$

3. 机械能: $E = E_k + E_p = \frac{1}{2}kA^2$ (简谐振动系统机械能守恒)

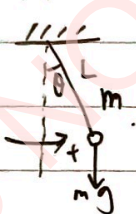
eg. 求 k .

$x_1 = x_2 = x$

$x = x_1 + x_2 = F(\frac{1}{k_1} + \frac{1}{k_2}) \quad k = k_1 + k_2$

$\rightarrow k = \frac{k_1 \cdot k_2}{k_1 + k_2}$

eg. 单摆.



$M = -mg \sin\theta \approx -mgL\theta$

$M = J\alpha$

$J = mL^2$

$\Rightarrow -mgL\theta = mL^2 \frac{d^2\theta}{dt^2}$

$\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0 \rightarrow \omega = \sqrt{\frac{g}{L}}$

(eg.) 1. 简谐运动方程.

$mg - kb = 0$

$$\begin{cases} F_{\text{合}} = mg - k(b+x) \\ = -kx \end{cases}$$

$F_{\text{合}} = ma = m \frac{d^2x}{dt^2}$

$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$

$\Rightarrow \text{简谐} \quad \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{b}}$

$t=0 \text{ 时, } x_0 = -b, v_0 = 0 \Rightarrow A = b, \varphi = \pi$

$\Rightarrow x = b \cos(\sqrt{\frac{g}{b}}t + \pi)$

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三. 合成

1. 同向同频合成

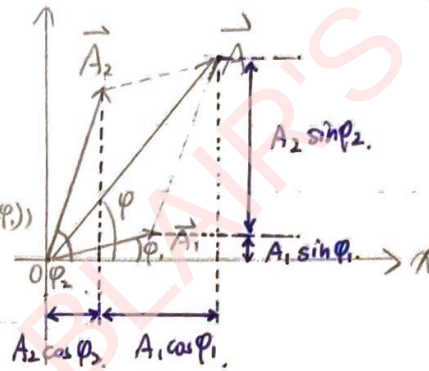
$$\begin{cases} x_1 = A_1 \cos(\omega t + \varphi_1) \\ x_2 = A_2 \cos(\omega t + \varphi_2) \end{cases}$$

$$A = A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)$$

利用余弦定理

$$A = A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)$$

$$\tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$$



2. 同向不同频合成 → 拍

$$\begin{cases} x_1 = A \cos \omega_1 t \\ x_2 = A \cos \omega_2 t \end{cases}$$

随 t 缓变
A(t)

随 t 快变
cos ωt

→ 合振动: 振幅缓变的简谐振动

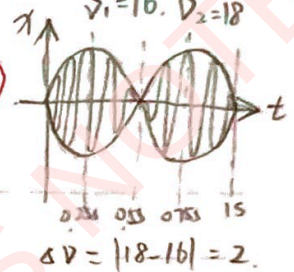
$$x = x_1 + x_2 = 2A \cos\left(\frac{\omega_2 - \omega_1}{2} t\right) \cos\left(\frac{\omega_2 + \omega_1}{2} t\right)$$

非简谐

当 $\omega_1 \sim \omega_2$ 时, $\omega_2 - \omega_1 \ll \omega_2 + \omega_1 \Rightarrow x = A(t) \cos \omega t$

Δ拍: 合振动忽强忽弱的现象

拍频: 单位时间, 强弱变化次数, $\Delta \nu = |\nu_2 - \nu_1|$



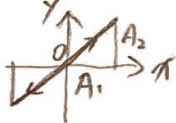
3. 垂直同频合成

$$\begin{cases} x = A_1 \cos(\omega t + \varphi_1) \\ y = A_2 \cos(\omega t + \varphi_2) \end{cases}$$

$$\text{合运动: } \frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - \frac{2xy}{A_1A_2} \cos(\varphi_2 - \varphi_1) = \sin^2(\varphi_2 - \varphi_1) \text{ 消参}$$

$$(1) \varphi_2 - \varphi_1 = 0$$

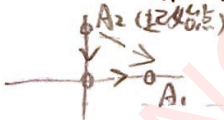
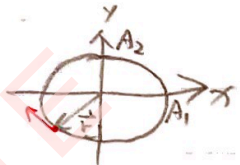
$$\Rightarrow y = \frac{A_2}{A_1} x$$



$$A = \sqrt{A_1^2 + A_2^2}$$

$$(2) \varphi_2 - \varphi_1 = \frac{\pi}{2}$$

$$\Rightarrow \frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} = 1$$



顺时针 (右旋)

$$\Delta \varphi \quad 0 \quad \frac{\pi}{4} \quad \frac{\pi}{2} \quad \frac{3\pi}{4} \quad \pi \quad \frac{3\pi}{2} \quad 2\pi$$

合成图



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4. 垂直不同频合成.

$$\Delta\varphi = (\omega_2 - \omega_1)t + (\varphi_2 - \varphi_1)$$

if $\omega_2 \sim \omega_1$, 可视为同频, 但 $(\varphi_2 - \varphi_1)$ 随 t 缓变, 即按上页合成图顺序缓变.

Δ if 2 振动频率成整数比 $\rightarrow$ 轨迹为李萨如图. ($\omega_1 : \omega_2 = a_2 : a_1$)

切点处.

第四章. 波动 (wave)

No.

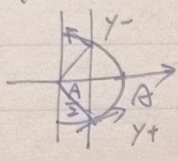
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一. 波动方程. ($y(x, t)$)

$$\begin{aligned} y(x, t) &= A \cos[\omega(t \pm \frac{x}{u}) + \varphi] \\ &= A \cos[2\pi(\frac{t}{T} \pm \frac{x}{\lambda}) + \varphi] \\ &= A \cos(\omega t - kx + \varphi). \quad k = \frac{2\pi}{\lambda}. \end{aligned}$$

eg. 1. 一平面简谐波, 向 x 轴负方向传播, $u = 120 \text{ m/s}$, $\lambda = 60 \text{ m}$. 原点处 $y = \frac{A}{2}$ 并向 y_+ 运动作为计时 0 点, 求波动方程.

$\checkmark$ $\begin{aligned} u &= 120 \text{ m/s} \\ \lambda &= 60 \text{ m} \end{aligned} \rightarrow v = \frac{u}{\lambda} = 2 \text{ Hz} \rightarrow \omega = 2\pi v = 4\pi.$

 $\Rightarrow \varphi = -\frac{\pi}{3} \quad k = \frac{2\pi}{\lambda} = \frac{\pi}{30}$

$$y = A \cos(4\pi t + \frac{\pi}{30}x - \frac{\pi}{3})$$

x 为负.

二. 波的能量与强度.

$$y = A \cos \omega(t - \frac{x}{u})$$

$$v = -A\omega \sin \omega(t - \frac{x}{u})$$

$$\begin{aligned} dE_k &= \frac{1}{2} dm v^2 = \frac{1}{2} \rho dV A^2 \omega^2 \sin^2 \omega(t - \frac{x}{u}) \\ &= dE_p. \end{aligned}$$

$$dE = dE_k + dE_p = \rho dV A^2 \omega^2 \sin^2 \omega(t - \frac{x}{u})$$

(波的能量 = 振动动能 + 弹性势能)

能量密度: $w = \frac{dE}{dV} = \rho A^2 \omega^2 \sin^2 \omega(t - \frac{x}{u})$

平均能量密度: $\bar{w} = \frac{1}{T} \int_0^T w dt$

$$= \frac{1}{2} \rho A^2 \omega^2.$$

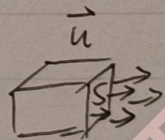
能流: $P = S w u$. 单位时间通过某一面积 S 的波能

平均能流: $\bar{P} = S \bar{w} u$. (一个周期内的能流平均值).

波的强度: $I = \bar{w} u = \frac{1}{2} \rho A^2 \omega^2 u$.

(能流密度)

$$I = \frac{P}{S} = \frac{P}{2\pi R^2}$$



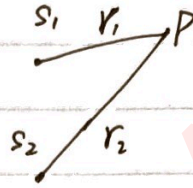
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三、波场强度分布.

$$y(x, t) = A \cos(\omega t - \beta x + \varphi)$$

1. 任一点合振动.

(设振动方向 $\perp$ 纸面).

$$S_1: \xi_{10} = A_{10} \cos(\omega t + \varphi_{10})$$

$$S_2: \xi_{20} = A_{20} \cos(\omega t + \varphi_{20})$$

• P点两分振动.

• P点合振动.

$$\xi_1 = A_1 \cos(\omega t + \varphi_{10} - \beta r_1)$$

$$\xi = \xi_1 + \xi_2 = A \cos(\omega t + \varphi)$$

$$\xi_2 = A_2 \cos(\omega t + \varphi_{20} - \beta r_2)$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \Delta \varphi}$$

$$\Delta \varphi = (\varphi_{20} - \varphi_{10}) - \beta(r_2 - r_1)$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Delta \varphi$$

2. 加强、减弱条件.

• 加强. $\Delta \varphi = \pm 2m\pi$. ($m=0, 1, 2, \dots$)• 减弱. $\Delta \varphi = \pm (2m+1)\pi$. ($m=0, 1, 2, \dots$)

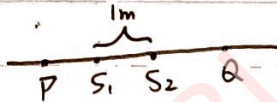
$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

$$I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

$$\text{if } A_1 = A_2, I_{\max} = 4I_1$$

$$\text{if } A_1 = A_2, I_{\min} = 0$$

eg. $A = 0.02 \text{ m}$. $u = 32 \text{ m/s}$.
 $\nu = 4 \text{ Hz}$. $\varphi_{10} = 0$. $\varphi_{20} = \frac{\pi}{4}$.

求 (1) S_1, S_2 振动方程. (2) 求 P, Q 振幅.

$$\checkmark \text{ 解: (1) } y_1 = 2 \cos 8\pi t. \quad y_2 = 2 \cos(8\pi t + \frac{\pi}{4}).$$

$$(2). \lambda = uT = 8 \text{ m}.$$

$$P: \Delta \varphi = (\varphi_2 - \varphi_1) + \frac{2\pi}{\lambda}(r_{S_1 P} - r_{S_2 P}) = \frac{\pi}{4} + \frac{2\pi}{8}(-1) = 0 \Rightarrow A = A_1 + A_2 = 4 \text{ cm}.$$

$$Q: \Delta \varphi = (\varphi_2 - \varphi_1) + \frac{2\pi}{\lambda}(r_{S_1 Q} - r_{S_2 Q}) = \frac{\pi}{4} + \frac{2\pi}{8}(+1) = \frac{\pi}{2} \Rightarrow A = \sqrt{A_1^2 + A_2^2} = 2\sqrt{2} \text{ cm}.$$

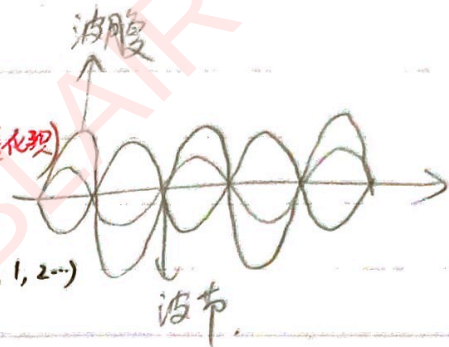
四. 驻波.

两列相干波 反向传播而叠加.

$$\xi_1 = A \cos(\omega t - \frac{\pi}{\lambda} x) \rightarrow x$$

$$\xi_2 = A \cos(\omega t + \frac{\pi}{\lambda} x) \leftarrow x.$$

$$\Rightarrow \xi = \xi_1 + \xi_2 = (2A \cos \frac{\pi}{\lambda} x) \cdot \cos \omega t. \quad (\text{和差化积})$$



• 波腹处: $|\cos \frac{\pi}{\lambda} x| = 1 \rightarrow x = \pm k \frac{\lambda}{2} \quad (k=0, 1, 2, \dots)$

• 波节处: $|\cos \frac{\pi}{\lambda} x| = 0 \rightarrow x = \pm (2k+1) \frac{\lambda}{4}$

△相位: 无相位传播.

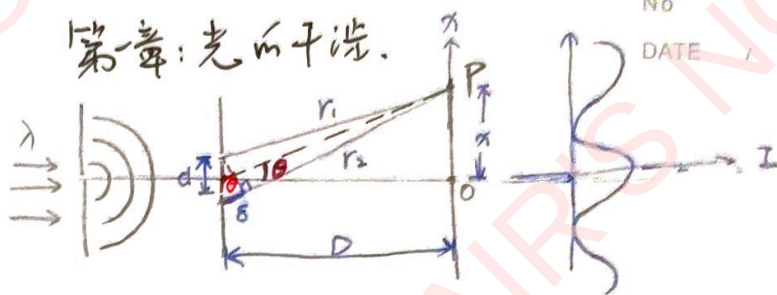
△能量: 无能量传播.

△相位差: 同一分段同相, 相邻分段反相.

五. 半波损失. (波疏 $\rightarrow$ 波密)

第一章：光的干涉。

1. 双缝干涉。



干涉特点：

(1) 平行、明暗相间条纹。

$$d \gg \lambda, D \gg d.$$

(2) θ 不太大时，条纹间距 $\delta = r_2 - r_1 \approx d \sin \theta \approx d \tan \theta = d \frac{x}{D}$

$$\Delta \varphi = \frac{2\pi}{\lambda} \cdot \delta$$

$\Rightarrow$ 明纹： $\Delta \varphi = \pm k\lambda$ $k = 0, 1, 2, \dots$

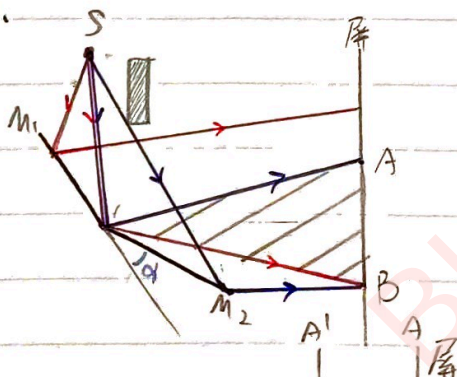
暗纹： $\Delta \varphi = \pm (2k+1)\lambda$

$$\Delta x = \frac{D}{d} \cdot \lambda$$

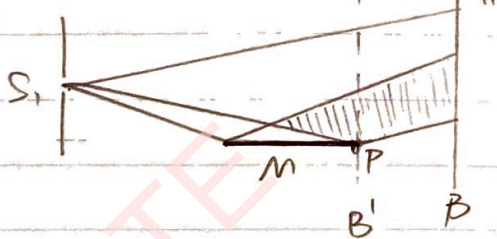
两明纹间距。

$$\Delta \text{光强公式：} I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Delta \varphi.$$

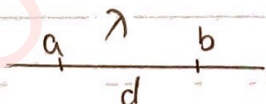
2. 菲涅耳双镜。



3. 劳埃镜。



4. 光程。



in vacuum: $\Delta \varphi = \varphi_b - \varphi_a = \frac{2\pi}{\lambda} \cdot d$

in the medium: $\Delta \varphi = \frac{2\pi}{\lambda_n} \cdot d$ $\lambda_n = \frac{u}{\lambda} = \frac{c/n}{\lambda} = \frac{c}{\lambda n} = \frac{\lambda}{n}$

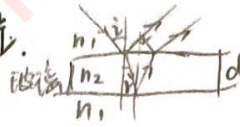
$$\Rightarrow \Delta \varphi = \frac{2\pi n d}{\lambda}$$

光程： $L = nd$

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① 牛顿干涉.



$$\delta = 2d \sqrt{n_2^2 - n_1^2 \sin^2 \theta} + \frac{\lambda}{2} \begin{cases} = R\lambda & \text{加强} \\ = (2R+1)\frac{\lambda}{2} & \text{减弱} \end{cases}$$

5. 薄膜干涉. ② 等厚干涉.

$$\text{暗纹: } \delta = 2ne_R + \frac{\lambda}{2} = (2R+1)\frac{\lambda}{2}$$

$$\Rightarrow e_R = \frac{R\lambda}{2n}$$

$$e_{R+1} - e_R = \frac{\lambda}{2n} = \frac{\lambda_n}{2}$$

$$= l \sin \theta \Rightarrow l = \frac{\lambda_n}{2n \sin \theta} \quad (\text{劈尖公式})$$

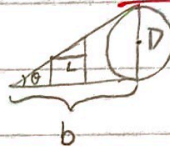
$$(n = \frac{c}{v}, \text{ 折射率 } v \downarrow, \lambda \downarrow)$$

$$\boxed{l = \frac{\lambda_n}{2\theta}} \quad \star$$

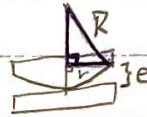
e.g. 已知 λ, b, n 条明纹间距 s , 求直径 D .

$$L = \frac{s}{n-1}$$

$$D = b \sin \theta \approx b \cdot \frac{\frac{\lambda}{2}}{L} = \frac{b\lambda(n-1)}{2s}$$



6. 牛顿环.



中间为 0 级暗纹.

$$R^2 = (R-e)^2 + r^2 \quad \uparrow \quad R^2 = R^2 - 2Re + e^2 + r^2$$

$$\Rightarrow r^2 = 2Re$$

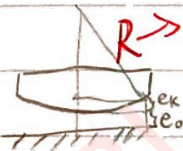
$$\delta = 2e + \frac{\lambda}{2} = \frac{r^2}{R} + \frac{\lambda}{2}$$

e.g. 求如图干涉实验第 k 级暗纹半径.

$$e_k = \frac{r_k^2}{2R}$$

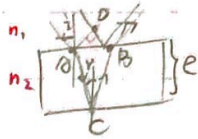
$$(2k+1)\frac{\lambda}{2} = 2(e_k + e_0) + \frac{\lambda}{2}$$

$$\Rightarrow r_k = \sqrt{R(k\lambda - 2e_0)} \quad (R \gg \frac{2e_0}{\lambda})$$

 $R \rightarrow$ 曲率半径.

7. 平行薄膜表面干涉.

$$\delta = 2AC \cdot n_2 + BD \cdot n_1 + \frac{\lambda}{2}$$



$$+ \begin{cases} \frac{\sin i}{\sin r} = \frac{n_2}{n_1} \\ \cos r = \frac{e}{AC} \\ \tan r = \frac{AB/2}{e} \\ \sin i = \frac{BD}{AB} \end{cases}$$

$$\Rightarrow \delta = 2e \sqrt{n_2^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2}$$

8. 迈克耳逊干涉仪.

e.g. 插片 - $n=1.54$ 透明介质. 移动 20 条纹. $\lambda = 5893 \text{ \AA}$.
求厚度 d .

$$2(n-1)d = 20\lambda$$

$$d = \frac{10\lambda}{n-1} \approx 10.9 \mu\text{m}.$$

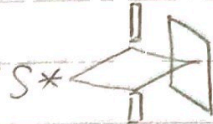
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第二章. 光的衍射.

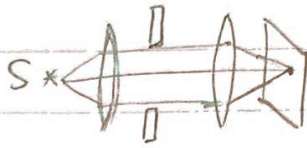
1. ① 费涅耳衍射.

(近场衍射)



② 夫朗和费衍射.

(远场衍射)



2. 惠更斯-菲涅耳原理.

(子波)

(子波相叠加决定衍射中各点复).

二. 单缝 的夫朗和费衍射. 半波带法.

$$\text{暗: } a \sin \theta = \pm 2k \cdot \frac{\lambda}{2}, \quad k=1, 2, 3, \dots$$

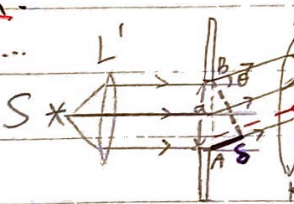
(偶数半波带)

$$\text{明: } a \sin \theta = \pm (2k+1) \frac{\lambda}{2}, \quad k=1, 2, 3, \dots$$

(奇数半波带)

$$\text{圆孔衍射: } \theta = 1.22 \frac{\lambda}{D}$$

(最小分辨角)



光强分布.

$$I \propto \text{振幅} E^2$$

$$I = I_0 \frac{\sin^2 u}{u^2}$$

$$(\text{明纹}) = I_0 \frac{\sin^2 \left(\frac{(2k+1)\pi}{2} \right)}{\left(\frac{(2k+1)\pi}{2} \right)^2}$$

三. 光栅.

$$\text{干涉明纹: } (a+b) \sin \theta = k\lambda$$

$$\text{衍射暗纹: } a \sin \theta' = k'\lambda$$

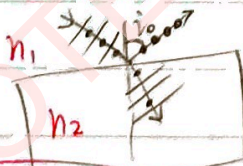
$$\frac{a+b}{a} = k \text{ 缺级}$$

四. 偏振.

线偏振光
部分...
圆...
自然光

$$1. \text{马吕斯定律: } I_2 = I_1 \cos^2 \alpha$$

$$2. \text{布儒斯特角: } n_1$$

起偏角: i_0

$$\tan i_0 = \frac{n_2}{n_1}$$